



SAINT PIERRE
LEARNING RESOURCES

MATHEMATICS

College Algebra

Linear Models in Context

College • Introductory Algebra

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Teacher Guide

Level: College • Introductory Algebra

Standard / objective: Course objective — Interpret slope and intercept, construct linear models, solve linear equations, and evaluate model limitations.

Source / course note: Course-level objective; no accreditation or universal equivalency claimed.

Before this lesson: Solve one-step equations and interpret ordered pairs.

Model and discuss: For $C(h)=18+6h$, identify the \$18 starting fee and \$6 per hour. Set $18+6h=72$, subtract 18, then divide by 6 to get 9 hours.

Watch for: The intercept is the starting cost, not the hourly rate. Units should appear in interpretations.

Suggested sequence: 5-minute launch, 10-minute model, 20-30 minutes of practice, then a 5-minute exit ticket. Use separate paper for extended writing.

Learning supports

AIG: Compare two pricing models and identify when each costs less.

504: Offer chunked tasks, reduced copying, extra time, and breaks as specified in the learner's plan.

EC: Use a table for $h=0,1,2$ before writing the rule; annotate units.

ESL: Rehearse: Each additional hour changes cost by _____. At zero hours, _____.

Adapt these suggestions to the current learning plan. Print on US Letter paper at actual size; keep the key separate. One teacher may use the resource with their own students; no public posting or resale.



Student Directions & Practice

Model constant-rate situations and interpret parameters with units.

Directions: Complete every item. Show work or cite evidence. For open responses, answer in complete sentences.

- 1.** Find slope through $(2,7)$ and $(6,19)$.
- 2.** Write the line with slope 4 and y-intercept -3 .
- 3.** Solve $5x-7=28$.
- 4.** A service costs \$18 plus \$6 per hour. Write $C(h)$.
- 5.** Interpret the intercept in $C(h)$.
- 6.** Find $C(7)$.
- 7.** At what h is $C=72$?
- 8.** Why might a linear model fail for very large h ?



Assessment / Exit Ticket

Work independently. Show your reasoning. Use extra paper if needed.

1. A rental costs \$12 plus \$4 per hour. Write $C(h)$, then find $C(5)$.
2. For this model, how many hours cost \$40? Explain the meaning of the solution.

Reflection: What helped you decide? What do you want to practice next?

Score each task: 2 = accurate with clear reasoning; 1 = partly correct or missing reasoning; 0 = incorrect or no evidence. The reflection is not scored.

Answer Key & Teaching Guidance

1. 3
2. $y=4x-3$
3. $x=7$
4. $C(h)=18+6h$
5. \$18 fixed starting fee.
6. \$60
7. 9 hours
8. Rates, capacity, discounts, or conditions may change outside the observed range.

Exit ticket guidance

1. $C(h)=12+4h$; $C(5)=$32$.
2. $12+4h=40$ gives $h=7$ hours. Seven hours of rental costs \$40 under this pricing rule.

Next teaching move: Use the worked model again if the concept is unclear. If the answer is correct but reasoning is incomplete, ask for an explanation. Offer the extension once both tasks are secure. Scores guide instruction; they are not standardized mastery cutoffs.